BAGEL · Frozen GEN + Spatial UND Velocity Correction

Method

❄ Frozen Trainable Frozen context

A BAGEL MoT · T2I

❄ **UND expert**

Prompt → embeddings

QKV

Multi-modal attention

prompt KV cached → GEN reads

FFN

prompt KV per layer

❄ **GEN expert**

vae2llm(x_t) + $T(t)$

QKV

Multi-modal attention

FFN

image features per layer · v_t

× 28

❄ frozen, reused below

$v_{\text{conditional}}$, v_{guided}

prompt KV (per layer)

image KV (per layer)

velocity v_t

CFG and the corrector read these states

B Training

x_0, ε

1 Sample

image x_0
caption
ε, t

$x_t = (1-t)x_0 + t\varepsilon$

2 Frozen GEN + CFG

conditional pass
reference pass

$v_{\text{conditional}}$
v_{guided}

$\hat{x} = x_t - t\,v_{\text{guided}}$

3 Target

$g = v_{\text{guided}} - v_{\text{conditional}}$

$H_g(z) = z - \min(\langle z, g\rangle / \|g\|^2, 0)\, g$

target = $H_g(\varepsilon - x_0 - v_{\text{conditional}})$

4 Fit

$r = \text{Corrector}(\hat{x})$

loss = MSE(r, target)

gradients reach the corrector

The target keeps the guidance-preserving residual.

Sampling applies the same projection to the prediction.

❄ frozen: GEN and UND backbone

trainable: spatial corrector

C Inference

❄ prompt + image KV

VAE → image

Frozen GEN + CFG

v_{native}
KV

Proposal

$\hat{x} = x_t - t\,v_{\text{native}}$

Corrector

$r \to H_g(r)$

r_{used}

Euler step

$x \leftarrow x - \Delta t\,(v_{\text{native}} + r_{\text{used}})$

repeat $t: 1 \to 0$ · 50 nodes · $\Delta t = t_i - t_{i+1}$

$v = v_{\text{native}} + r_{\text{used}}$

$r_{\text{used}} = H_g(r)$

same projection as training